Dataset Split Methodology for the Volatility Final Project

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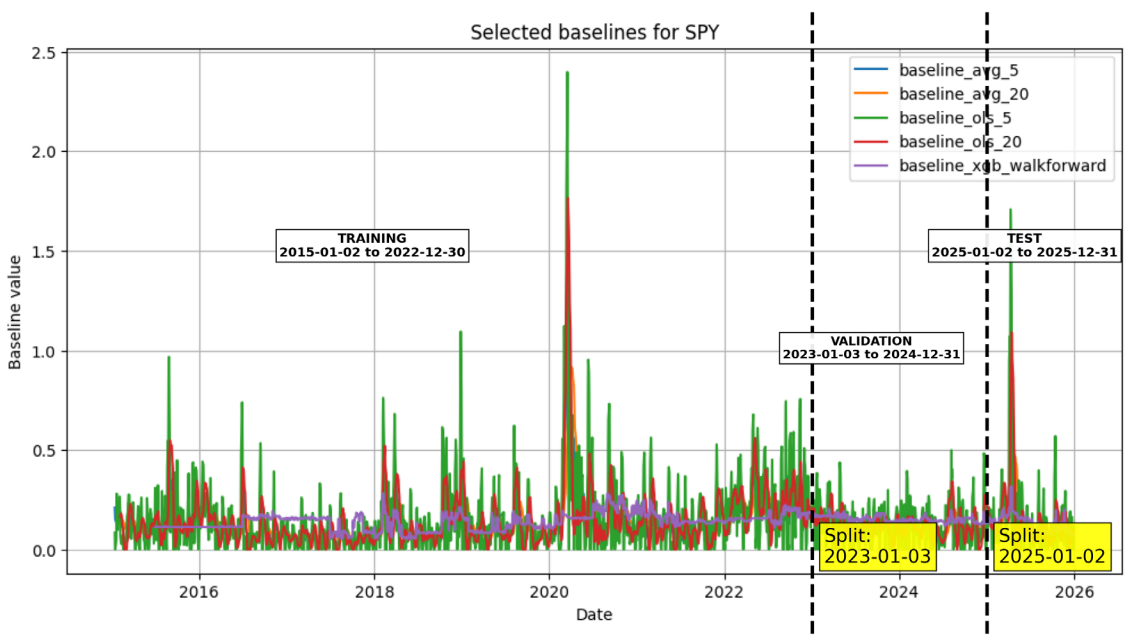
1. Summary

The dataset was split using a strict chronological time-based panel split rather than a random split. The chosen partitions were: Training (2015-01-02 to 2022-12-30), Validation (2023-01-03 to 2024-12-31), and Test (2025-01-02 to 2025-12-31).

The project predicts forward 5-trading-day realized volatility. Because the target depends on future returns, the split must preserve time order and avoid look-ahead leakage. A chronological split ensures that models are trained only on past information and evaluated on truly unseen future periods.

This structure gives models a long historical training period, a recent validation window for tuning, and a fully held-out final year for unbiased testing.



2. How the Dataset Was Split

The split was performed by date across all ETFs simultaneously rather than randomly assigning rows. The dataset is a panel dataset with many ETFs observed across the same trading dates, so the split must occur by date to maintain consistency across tickers.

Dataset partitions:

Training set: 2015-01-02 through 2022-12-30

Validation set: 2023-01-03 through 2024-12-31

Test set: 2025-01-02 through 2025-12-31

Resulting dataset proportions:

Training: 69,453 rows (71.95%)

Validation: 18,072 rows (18.72%)

Test: 9,000 rows (9.32%)

Total: 96,525 rows

3. Why the Split Had to Be Chronological

The prediction target is forward 5-trading-day annualized realized volatility. Each observation is indexed by (ticker, date), and the label is constructed from returns occurring in the five trading days following the observation date.

A random split would introduce look-ahead bias because training rows could indirectly contain information about future volatility regimes. A chronological split preserves the real-world prediction structure: using information available at time t to predict future volatility.

4. Criteria Followed in Designing the Split

Chronological ordering:

All splits respect strict time ordering. No random shuffling was used.

Consistent dataset partitions across models:

All baselines and advanced models use the exact same training, validation, and test rows so that performance comparisons reflect model differences rather than differences in data.

Respect for forward target structure:

Because the label depends on future data, observations near split boundaries require special treatment to avoid target leakage.

5. Leakage Control Rule

Rows near split boundaries may have targets that extend into the next partition. To prevent leakage, a five-trading-day purge gap is applied at each boundary.

Practical implementation:

- Do not train on the last five trading days of the training block.
- Do not evaluate on the last five trading days of the validation block.
- Remove rows whose target window crosses a partition boundary.

6. Why These Calendar Boundaries Are Appropriate

The boundaries align with natural calendar years and reflect meaningful market regimes.

Training period characteristics:

The training period includes calm periods, volatility spikes, the COVID shock, and the 2022 interest-rate tightening cycle.

Validation period characteristics:

The validation period reflects recent market conditions and is suitable for hyperparameter tuning.

Test period characteristics:

The test period represents a fully unseen market regime, allowing unbiased out-of-sample evaluation.

7. Importance for Team Model Comparison

For fair comparison across team models, all participants must use the same dataset splits, target definitions, purge rules, and evaluation metrics.

8. Evidence the Split Is Technically Sound

The split satisfies several integrity checks:

- Training maximum date < validation minimum date
- Validation maximum date < test minimum date
- No overlapping rows between partitions
- All rows accounted for within the dataset

9. Final Conclusion

The dataset split follows a chronological, leakage-aware methodology designed for forward volatility forecasting. The resulting proportions—approximately 72% training, 19% validation, and 9% test—are appropriate for financial time-series machine learning.

This structure provides a realistic modeling environment, ensures fair comparison across models, and preserves the integrity of out-of-sample testing.